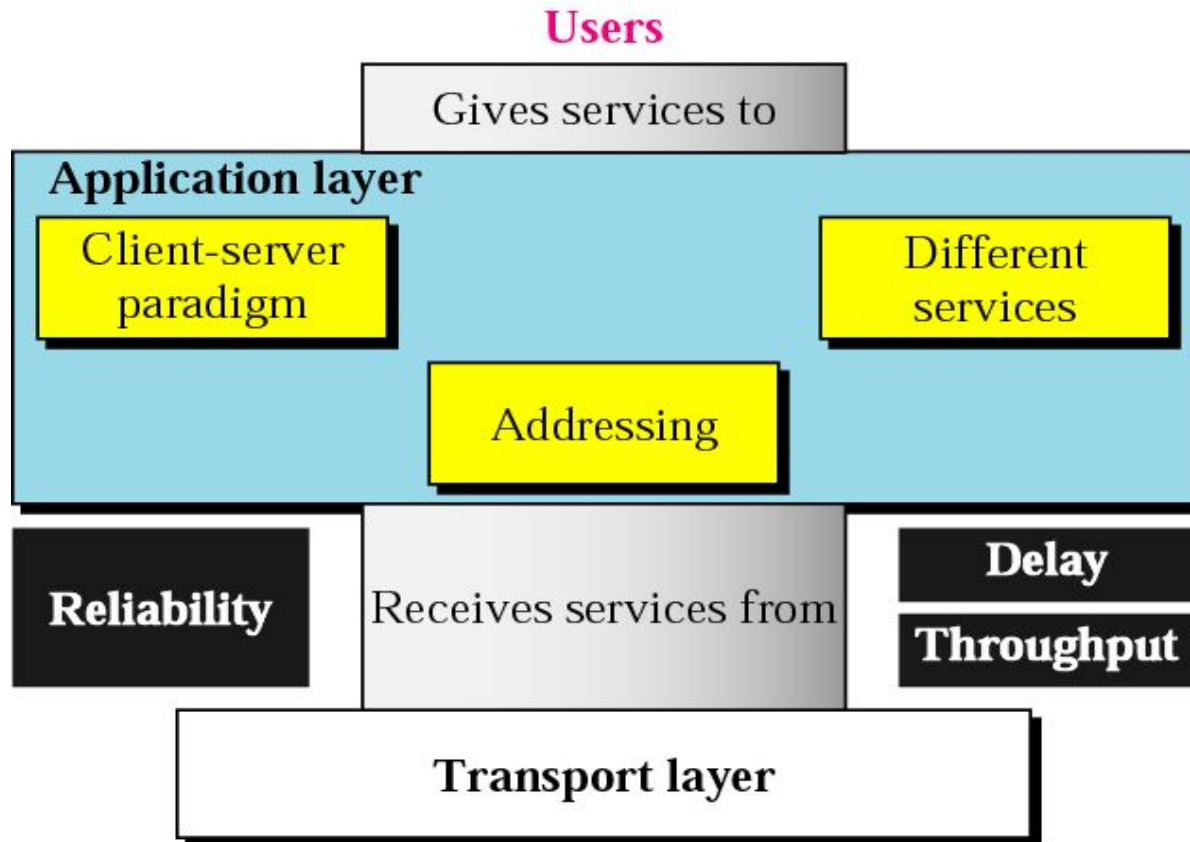


Computer Communication & Networks

Lecture 27-28

Application Layer: Domain Name System

Application Layer



Application Layer Topics to Cover

DNS

Remote Logging

Electronic Mail

File Transfer

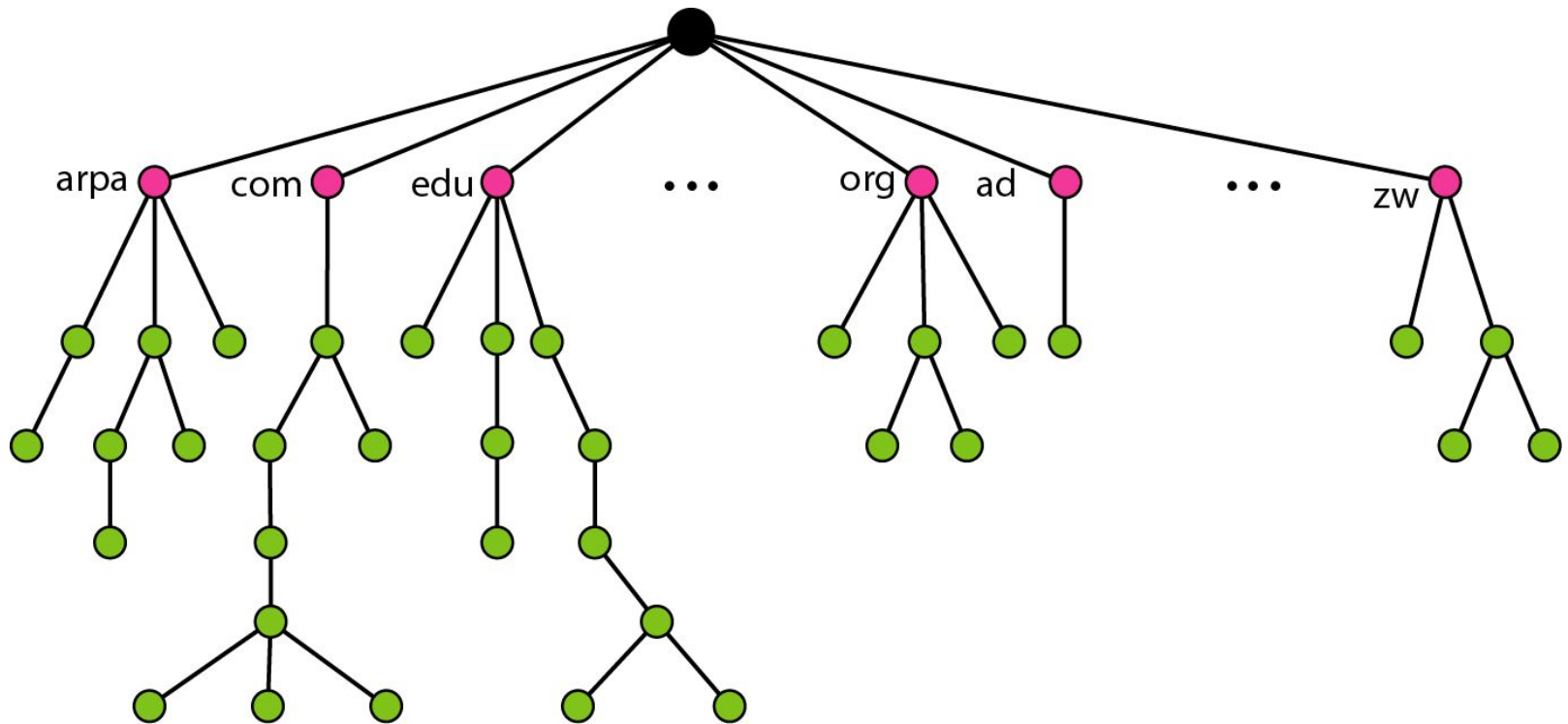
Name Space

- To be unambiguous, the names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses.

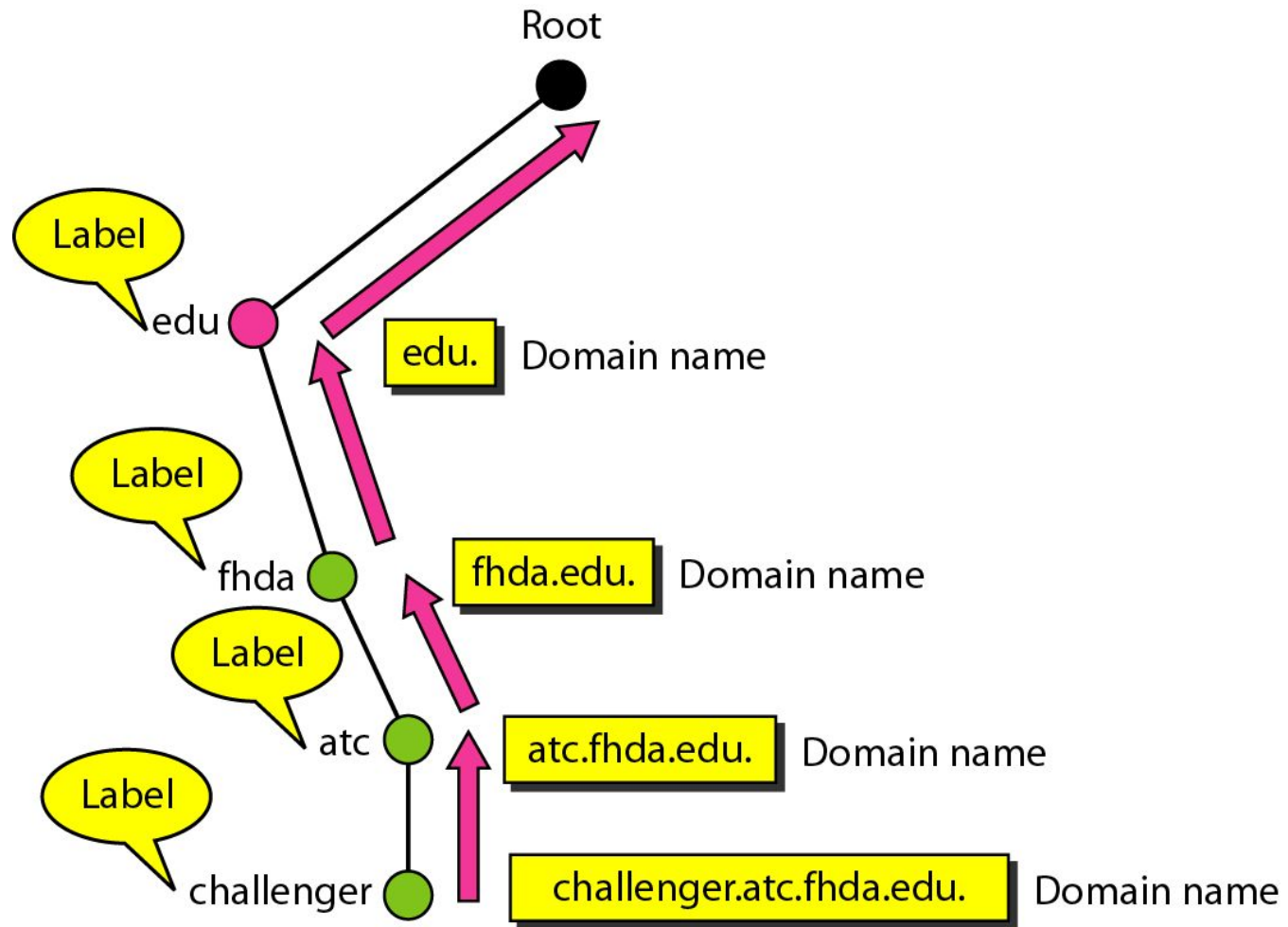
Domain Name Space

- To have a hierarchical name space, a domain name space was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127.

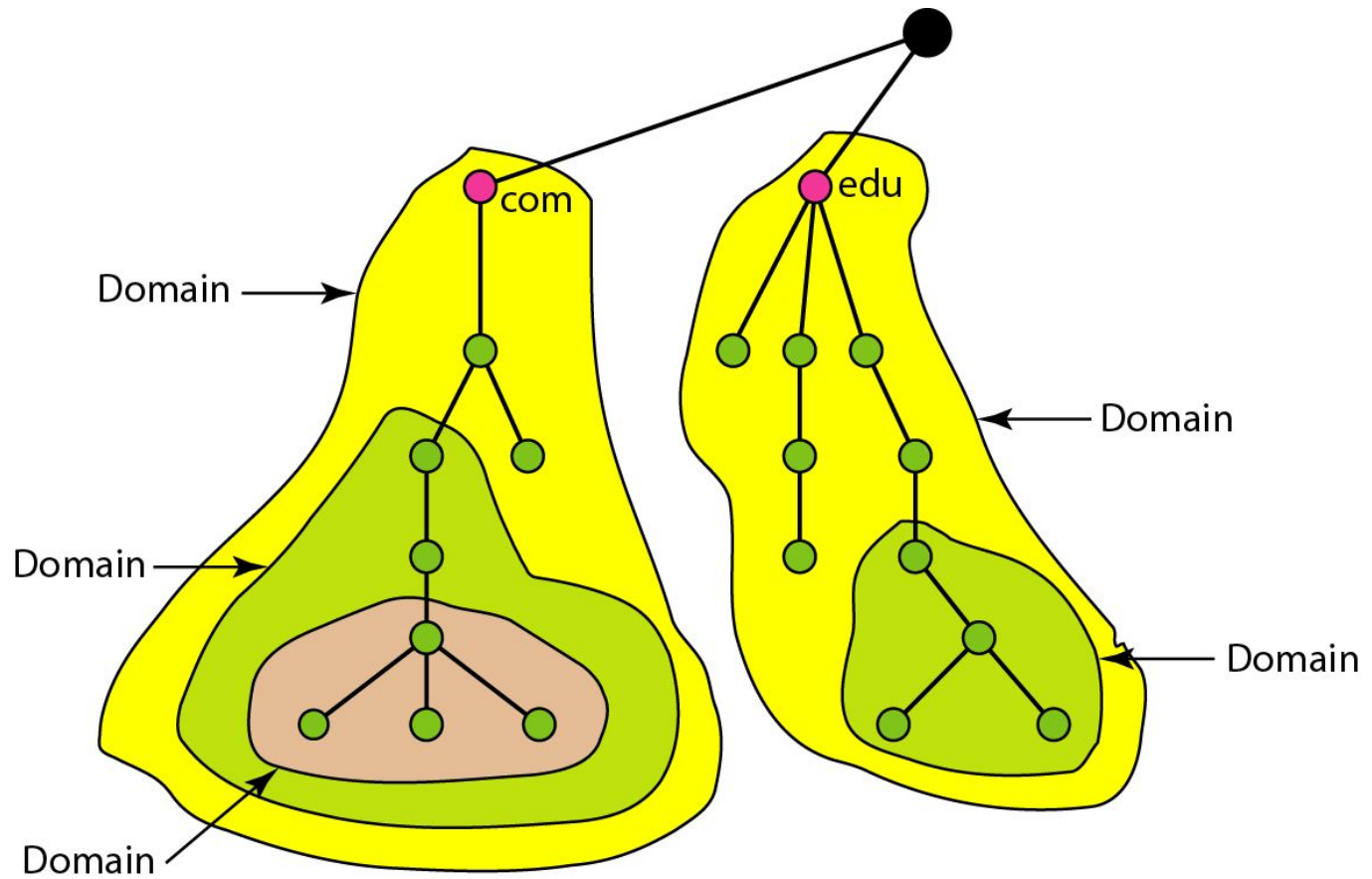
Domain name space



Domain names and labels



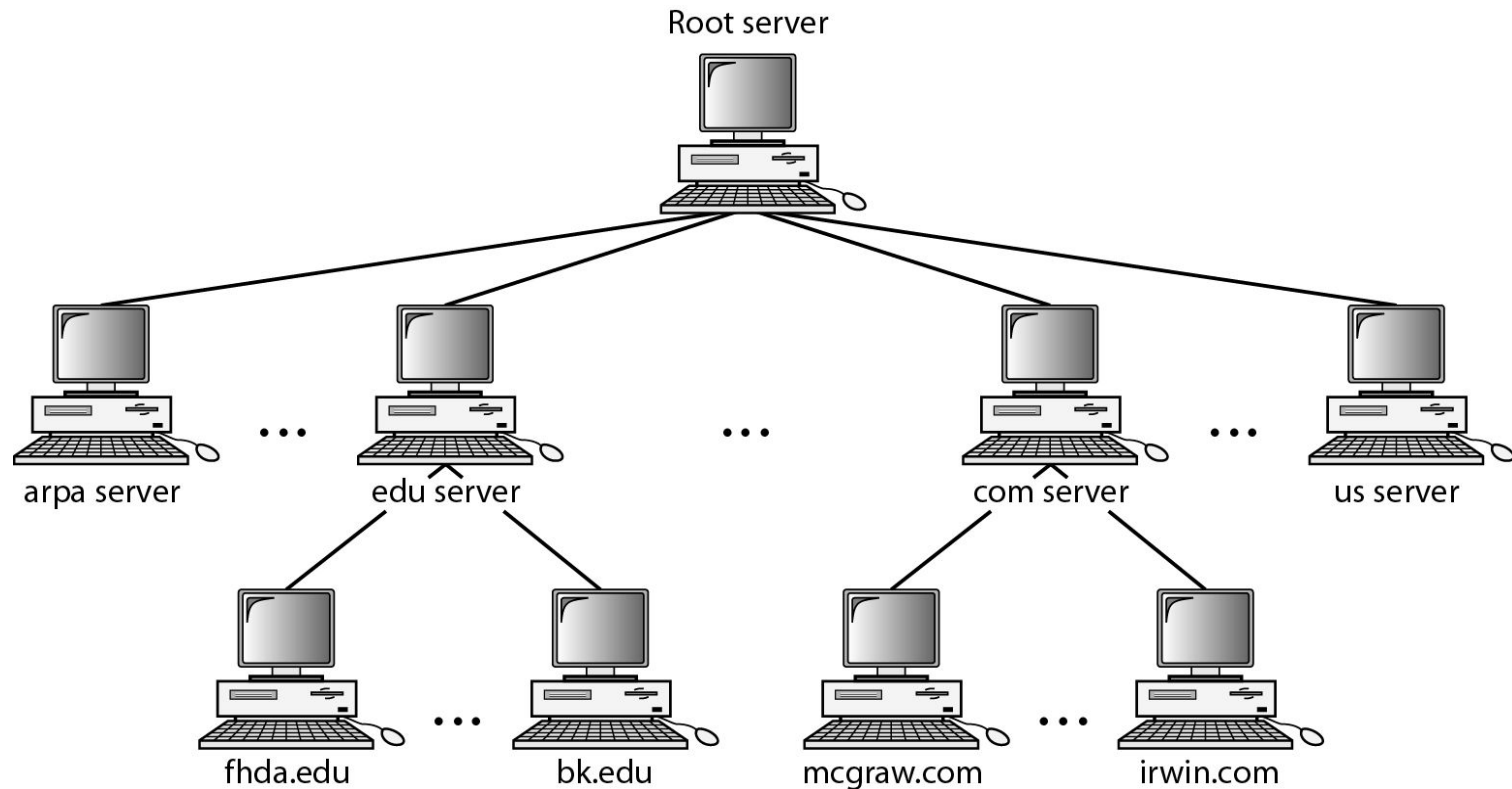
Domains



Distribution of Name Space

- The information contained in the domain name space must be stored. However, it is very inefficient and also unreliable to have just one computer store such a huge amount of information. In this section, we discuss the distribution of the domain name space.

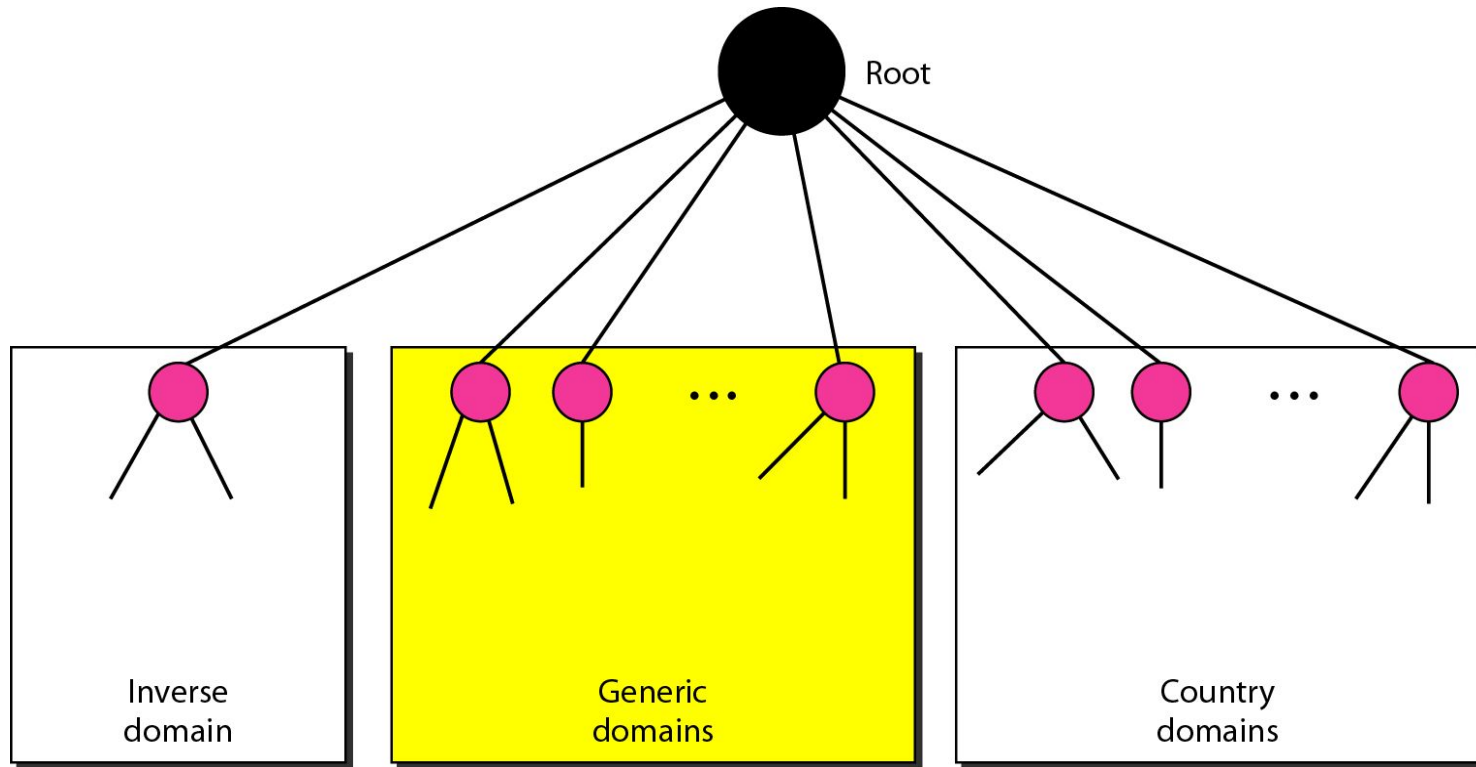
Hierarchy of Name Servers



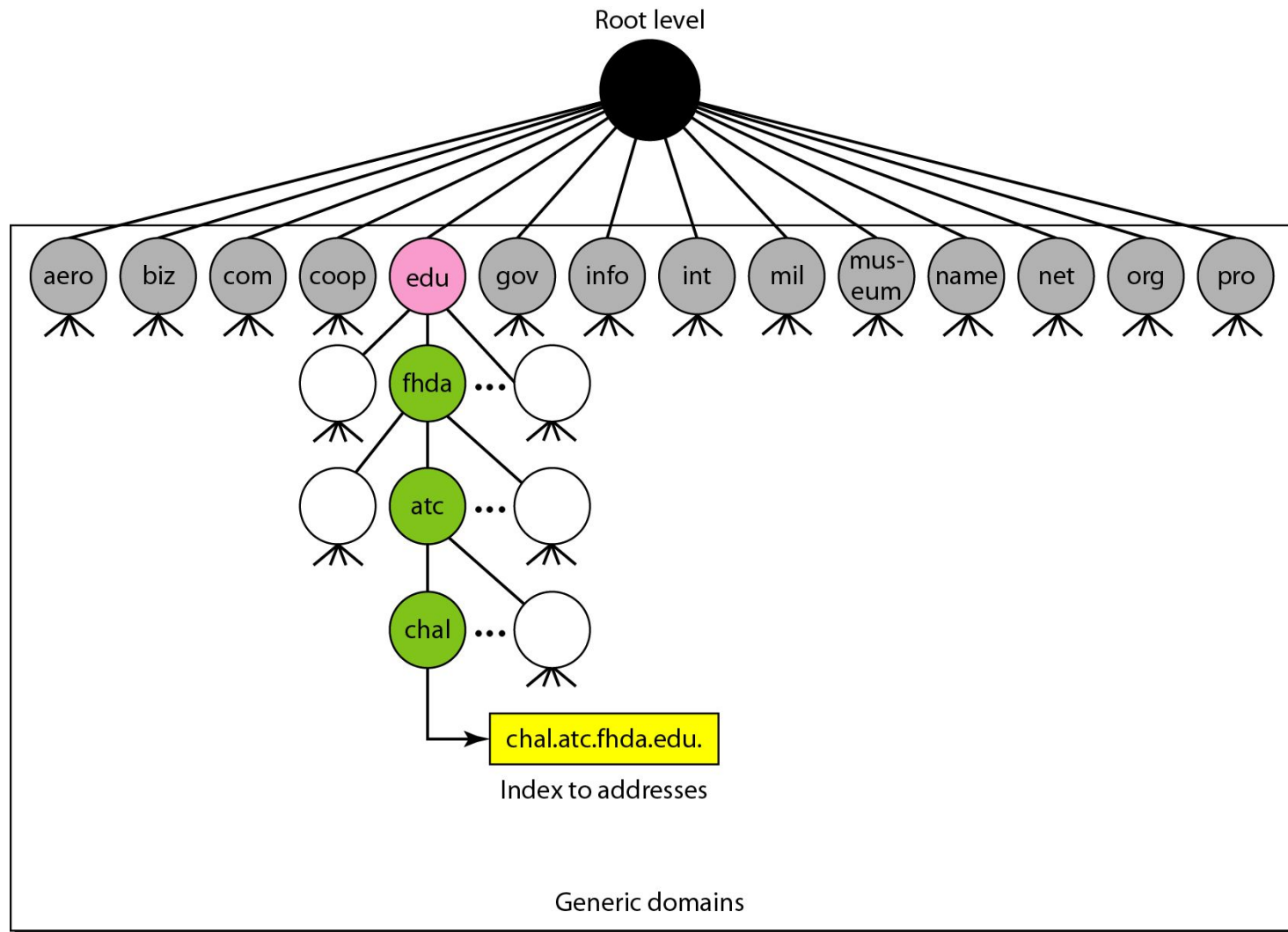
DNS in the Internet

- DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) is divided into three different sections: generic domains, country domains, and the inverse domain.

DNS IN THE INTERNET



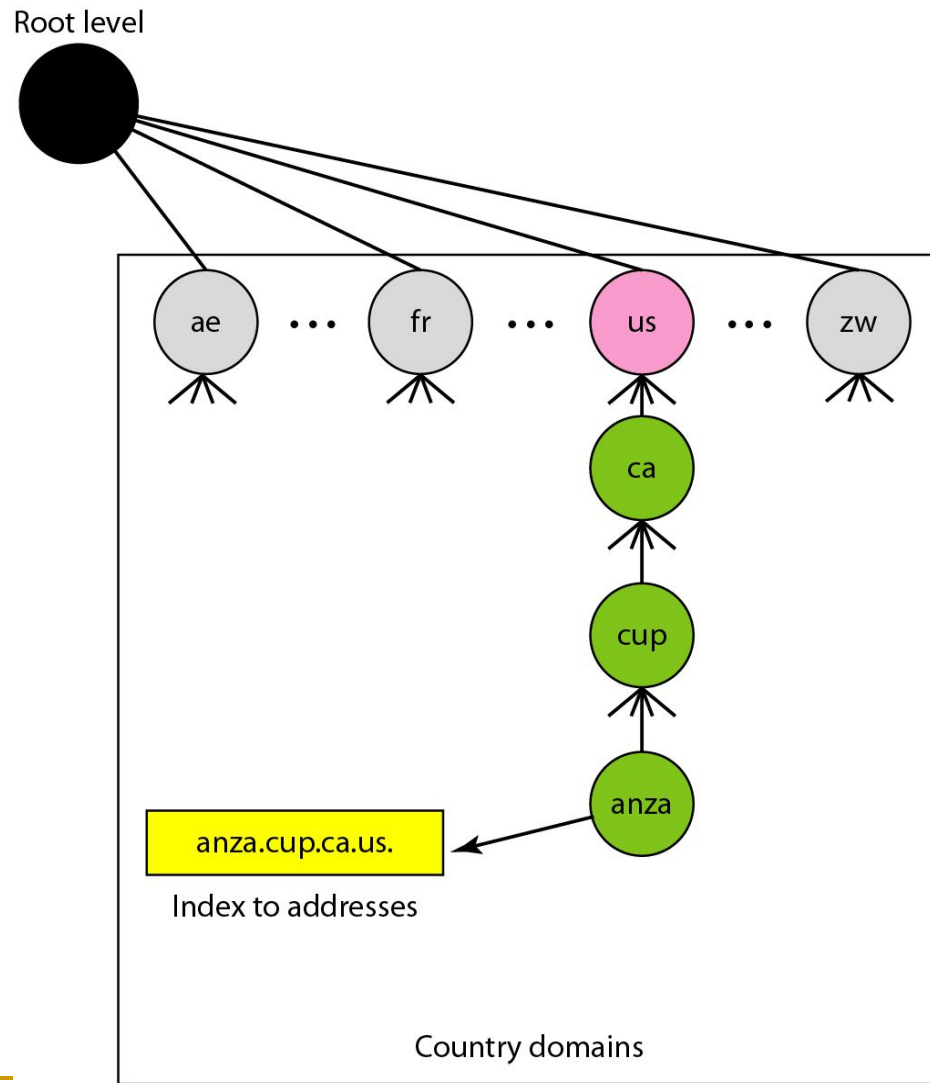
Generic Domains



Generic domain labels

<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

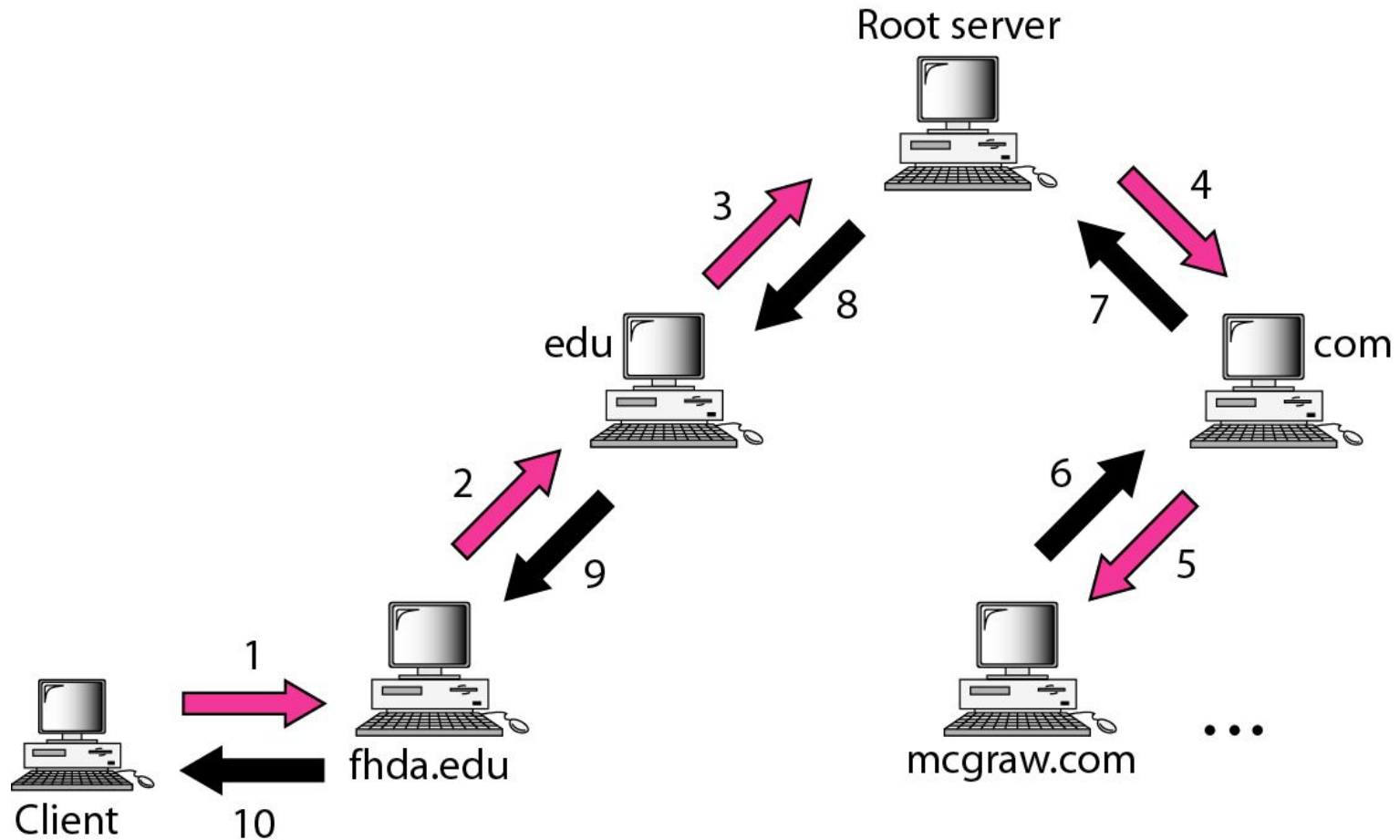
Country domains



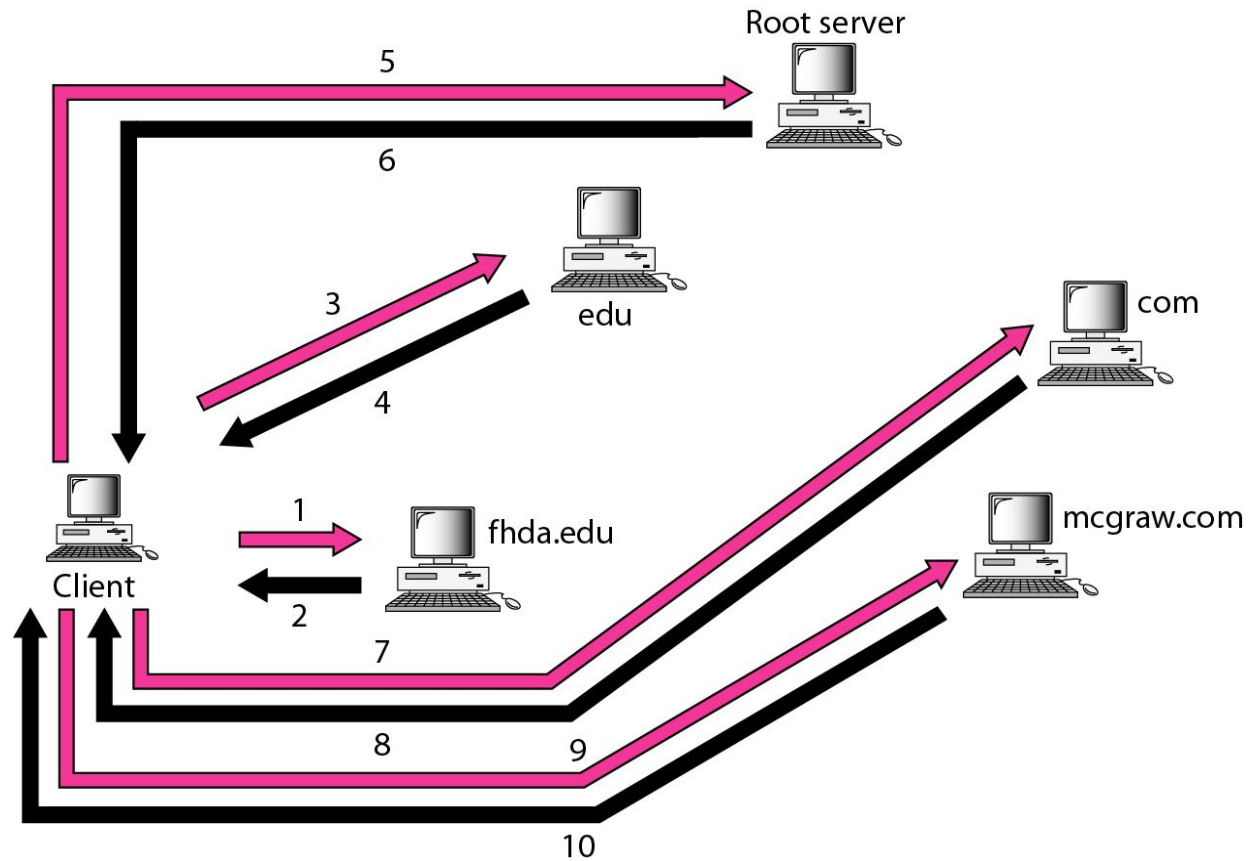
Resolution

- Mapping a name to an address or an address to a name is called name-address resolution.

Recursive resolution



Iterative Resolution



DNS: caching and updating records

- once (any) name server learns mapping, it *caches* mapping
 - cache entries timeout (disappear) after some time